Appendix D

Background on topological spaces

For the examples in this book all you really need to know is that topological spaces are sets with some extra structure enabling us to define continuous functions, that continuous functions are ones that do not break things apart, and that a homotopy is a continuous deformation of one continuous map into another. Here is a little more detail.

D.1 Open sets

A topological space is essentially a set equipped with a notion of "closeness", but that sounds like it has something to do with things being near each other, when it really doesn't. The technical definition can be thought of as a generalization of the idea of open intervals of the real line. Open intervals are the ones that do not contain the endpoints. For example, this is the open interval from 0 to 1: $\boxed{(0,1) = \{x \in \mathbb{R} \mid 0 < x < 1\}.}$

Open intervals are used all over the place in analysis, and if we combine them we get open sets. So we could have a disjoint union of two open intervals, say $\boxed{(0,1) \sqcup (10,11)}$ and it still counts as open as it doesn't have any of its endpoints (the formal definition is more precise than that). A non-disjoint union of open intervals is just another open interval; this is important as we often patch together properties like continuity across large open intervals by patching together small ones.

Here's an example of a non-disjoint union $\boxed{(0,2) \cup (1,3) = (0,3).}$ It has an overlap from 1 to 2. It's also rather useful that the overlap (that is, intersection) is itself another open interval $\boxed{(0,2) \cap (1,3) = (1,2).}$

This idea is then generalized to sets other than the real numbers. It's done in a way that is typical in abstract mathematics: not by finding some characterization of what "open" really means, but by examining some of the relationships that open sets satisfy on the real line, and then defining open sets by those behavioral properties rather than by any intrinsic ones. So the idea is that we have some other set X and we want to say what it means to have a collection of subsets of X called "open". Those key properties turn out to be these:

1. The empty set and the whole set X count as open.
2. Any union of open sets counts as open.
3. Any *finite* intersection of open sets counts as open.[†]

Note that the whole set counts as open, so if our set X is a closed interval, say $[0, 1]$, then although X is closed as a subset of $\mathbb{R}$, it is open as a subset of itself. Other subsets of X might be open in X although they're not open in $\mathbb{R}$, such as the half-open interval $[0, \frac{1}{2})$. This is open in $[0, 1]$ because the only closed end is the boundary (also we can check that the complement is closed) but it is only half-open in $\mathbb{R}$. We will need this in a moment.

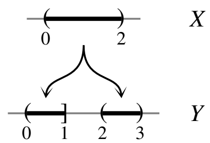
A system of open sets on X satisfying these behaviors is called a *topology* on X, and makes X into a *topological space*. It is possible to have different topologies on the same set.

D.2 Continuous functions

Arguably the point of having a topology is to define continuous functions, and these are sort of defined as "preserving closeness", but we have to be careful how we say it. It is tempting to say "open sets are preserved" but that's not quite right.

Consider this example of a function $f\colon X \longrightarrow Y$ which "breaks" the interval $(0, 2)$ in two. We're taking $X = (0, 2)$ and $Y = (0, 1] \sqcup (2, 3)$. A formal definition of f and an intuitive picture are shown below.

$$f(x) = \begin{cases} x & 0 < x \le 1 \\ x + 1 & 1 < x < 2 \end{cases}$$



Intuitively this shouldn't count as continuous because the interval X is broken apart at the point 1. However, this will not be detected if we think about open sets being preserved: an open interval in X might be split apart by f when it is mapped to Y, but the image will still be open in Y even though it's not open in $\mathbb{R}$. Instead we have to look at pre-images: we take open sets in Y and look at the set of all points in X that are mapped there. Now $(0, 1]$ is open in Y (as

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[†] We have to specify that we only care about finite intersections because infinite ones can turn out to be closed, for example: $\bigcap_n (-\frac{1}{n}, \frac{1}{n}) = [0, 0]$.

the closed end is just a boundary) but its pre-image in X is $(0, 1]$ which is not open in X, because the closed end is *not* a boundary in X. This is not formally stated, but captures the intuition.

So the definition of continuous function is as follows.

Definition D.1 Let X and Y be topological spaces. A function $f : X \longrightarrow Y$ is *continuous* if the pre-image of any open set in Y is open in X.

If you have studied formal real analysis you might want to investigate why this is equivalent to the epsilon–delta definition in the reals:

$$\forall a \in X \;\; \forall \varepsilon > 0 \;\; \exists\, \delta > 0 \quad \text{such that} \quad \left| x - a \right| < \delta \implies \left| f(x) - f(a) \right| < \varepsilon$$

The advantage of the definition using open sets is that it works in places where we are not using a notion of distance to define our topology. In the reals we are using open intervals to define the topology, and the open intervals are defined using distance: in fact this is the notion of metric space, a very particular type of topology.

D.3 Homotopy

A homotopy is a "continuous transformation" between continuous maps. The definition is very clever in that it doesn't require any extra levels of definition — the next dimension up still just uses continuous maps. The extra dimension is provided by making a sort of cylinder on X in the form of a product $I \times X$, rather than by inventing a new notion of map. Explaining this fully is somewhat beyond the scope of this appendix, but basically a homotopy from f to g is defined as a continuous map $\boxed{\alpha : I \times X \longrightarrow Y}$ such that $\alpha(0, x) = f(x)$ and $\alpha(1, x) = g(x)$. Here I is the (closed) unit interval $[0, 1]$ which we can think of as a unit of time. The product space $I \times X$ is like a cylinder on X where we have thickened it up, but at any time t (or cross-section) we have an entire copy of X. At time 0 we do the function f and at time 1 we do the function g and at every time in between we do something that gradually deforms our function from f into g.

This definition can essentially then be iterated by multiplying by I repeatedly to add more dimensions and get higher and higher dimensions of homotopy.

Glossary

This glossary is not to give formal definitions, but to give reminders to jog your memory about what something is.

2-category 2-dimensional category (with 0-cells, 1-cells and 2-cells).

Abuse of notation When we have used the same notation for two different things. Usually we do this when the things are in some sense the same and it would be somewhat tedious to distinguish between them with notation.

A priori In advance.

Arrow Another word for morphism in a category.

Bicategory Weak 2-dimensional category.

Bijection or bijective function Function that is both injective and surjective, that is, a "perfect matching" between inputs and outputs.

Binary operation A process that takes two inputs and produces one output.

Cat The category of all small categories and functors between them.

CAT The category of locally small categories and functors between them.

Categorical In the spirit of category theory. (Not to be confused with the way "categorical" is used to mean "very clear" in normal English.)

Co- Prefix indicating the dual of something (so all the arrows are reversed).

Commutative diagram A diagram of morphisms in which any paths of morphisms between the same endpoints yields the same composite.

Componentwise When you calculate something component by component.

Compose Do one thing and then another; in category theory we are usually "going along" one morphism and then another.

Cone (over a diagram) The basic data for a limit, consisting of an object (the vertex) and morphisms from the vertex to every object in the base diagram, such that everything in sight commutes.

Contravariant functor A functor from a dual category, that is, it reverses the direction of morphisms.

Covariant functor A functor that doesn't reverse the direction of morphisms. Usually we just say "functor" but sometimes we want to stress that it isn't contravariant, especially if there are some contravariant ones hanging around as is often the case when we're dealing with Yoneda.

Degenerate This is often used to mean that something is the identity that didn't need to be. For example a degenerate commutative square has some sides being the identity.

Degenerate category A category with only one object.

Discrete category A category with no arrows except identity arrows.

Disjoint union A disjoint union of two (or more) sets is taken by putting all the elements of all the sets in, but counting them all as different even if there appears to be an intersection. Informally we can think of it as painting each set's elements a different color to force them to be different.

Dynamically Reading a diagram "dynamically" means we progress through it region by region, reading each commuting region as an equality that takes us from one side to the other.

Dual category The same category but with the arrows considered to be pointing the other way. We write the dual of a category $\mathcal{C}$ as $\mathcal{C}^{\mathrm{op}}$.

Epic or epimorphism A categorical version of surjective function.

Equivalence of categories Good notion of sameness for categories, weaker than isomorphism.

Equivalence relation A relation that is reflexive, symmetric, and transitive.

Faithful functor A functor that is "locally injective" (on homsets).

Forgetful functor A functor that just forgets structure.

Free functor A functor that creates structure "freely", without any extra imposed equations.

Full functor A functor that is "locally surjective" (on homsets).

Functor Structure-preserving map between categories.

Functoriality The properties required of a functor: that it preserves identities and composition.

Graph Aside from the usual concept of a graph, in category theory this could refer to the underlying data for a category, consisting of a diagram of sets and functions like this: $C_1 \underset{t}{\overset{s}{\rightrightarrows}} C_0$

Group A monoid in which every element has an inverse. That is, a set with a binary operation that is associative and unital, and with respect to which every element has an inverse.

Groupoid A category in which every arrow has an inverse. A groupoid with one object is a group.

Homomorphism Generic word for structure-preserving map between structures. We can specify the context by saying things like "group homomorphism", "monoid homomorphism" and so on.

Homotopy A notion of map between continuous maps of topological spaces. It can be thought of as one dimension higher, and is essentially a continuous deformation of continuous maps.

Homset or hom-set General term for sets of morphisms in a (locally small) category.

Hypothesis This can either mean something you think is true that you haven't proved yet, or else it means one of the basic assumptions you've taken as the set-up for your theorem. Then when you prove it you say "by hypothesis" when you invoke that assumption.

Identity This can refer to an identity element with respect to a binary operation or composition, which is an element which "does nothing". It can also refer to an axiom, which might be called an identity if it identifies one thing with another in an equation.

iff if and only if

Image Given a function $f\colon A \longrightarrow B$, we can consider a particular element $a \in A$ and then we call $f(a)$ the image of a (under f). But we also say the image of A for the subset of B consisting of everything "hit" by f. To tell which type of image we mean, you need to type-check.

Injective function "No output is hit more than once."

Integers All the whole numbers, including positive numbers, negative numbers, and 0. Denoted $\mathbb{Z}$.

Inverse An element or morphism which "undoes" another one.

Invertible A morphism with an inverse.

Isomorphism Another word for an invertible morphism, perhaps with more emphasis on the source and target objects which are thus isomorphic, which is a more subtle notion of sameness in categories.

Lemma Small result that we prove, usually as a build-up to a larger theorem. Often lemmas sort out some technical details that we need for the theorem.

Locally small category A category $\mathcal{C}$ in which every collection of morphisms $\mathcal{C}(a, b)$ is in fact a set.

Modular arithmetic Arithmetic on an “n-hour clock”.

Monic or monomorphism Categorical version of injective function.

Monoid A category with only one object; that is, a set with a binary operation that is unital and associative.

Monoidal category Category with a binary operation on it (on the objects and also on the morphisms), often denoted $\otimes$.

Natural numbers The positive whole numbers, possibly including 0. Denoted $\mathbb{N}$.

Natural isomorphism An invertible natural transformation, that is, an isomorphism in a functor category.

Natural transformation A notion of morphism between functors.

Naturality Commuting conditions in the definition of natural transformation.

Non-image (Not formal terminology) Given a function $f\colon A \longrightarrow B$ I said “non-image” for the subset of B consisting of everything *not* hit by f.

Parallel Morphisms in a category are called parallel if they have the same source and the same target as each other.

Partially ordered set (poset) A category in which any pair of objects has at most one arrow between them.

Post-compose Compose a morphism after another one.

Pre-compose Compose a morphism before another one.

Preserves A functor preserves a certain structure if the structure still has the relevant properties after the functor is applied.

Presheaf/presheaves A presheaf on a category $\mathcal{C}$ is a functor $\mathcal{C}^{\text{op}} \longrightarrow \mathbf{Set}$.

Proposition Medium-sized result that we prove.

Pseudo- General prefix to indicate that something is up to isomorphism.

Putative We use this word when we’re studying a particular sort of thing, which I’ll call X. A putative X is something we are suggesting could be an X but we don’t know yet, because we haven’t checked it. Usually we say “putative X” when we’re about to check that it really is an X.

Quantifier A logical clause that tells us the scope of a statement. The usual quantifiers are “for all” and “there exists”.

Quintessential arrow category Informal terminology to describe a category consisting of just a single arrow. Similarly quintessential commuting square, quintessential isomorphism, and so on.

Rational numbers Fractions $\frac{a}{b}$ with a and b integers and $b \neq 0$. Denoted $\mathbb{Q}$.

Real numbers The rational numbers and irrational numbers together, making up a continuum of numbers on a line, with no gaps. Denoted $\mathbb{R}$.

"Recall" "I said this earlier and I'm hoping you have some recollection of it but if you don't there's a chapter or section for you to refer to."

Russell's paradox Paradox about self-reference in set theory, alerting us to the need for a hierarchy of concepts of "set" to avoid the self-reference.

Serially commutes This is when a diagram has some matching parallel arrows in it, and only the sub-diagrams of corresponding arrows commute.

Small category Category in which the collection of objects and the collection of morphisms are both sets.

Source The object at the "beginning" of a morphism in a category.

Strict Generally used to describe a structure in which axioms hold as equalities rather than isomorphisms.

Structure isomorphisms Structure in a higher-dimensional category coming from weakening something that was an axiom (via an equality) in a lower-dimensional version.

Sub- Prefix indicating a part of the whole (although the part could itself be all of it). Thus a subset of A is a set that is part of A (and could be all of it). Similarly subcategories. I also say sub-diagram informally.

Schematic diagram A diagram showing a general scheme of things rather than a formal diagram in a category.

Surjective function "Every output is hit at least once".

Target The object at the "end" of a morphism in a category.

Theorem A result with a proof.

Totally ordered set (toset) A category in which any pair of objects has exactly one arrow between them.

Type-checking A process of checking that all the concepts in a situation (or symbols in a formula) fit together in a way that makes "grammatical" sense, regardless of whether the result is true logically or not.

Underscore The symbol _ that we often use as a deliberate gap left for us to place a variable.

Unital A binary operation is called unital if it has a unit, that is, an object such that doing the binary operation with it does nothing. A composition is called unital if it has identities for every object, such that composition with them does nothing.

Vacuously satisfied A condition that is satisfied by virtue of it quantifying over the empty set. That is, we say "all such-and-such…" but that happens to be true because there aren't actually any such-and-suches.

Walking Another (informal) term for a "quintessential" category containing a certain structure. For example the "walking arrow" category just consists of two distinct objects and an arrow between them.

Weak Generally used to describe a structure in which axioms hold as isomorphisms or equivalences rather than equalities.

Well-defined Defined with good logic, without unresolved ambiguity.

Notation

$\forall$	for all
$\exists$	there exists
!	unique, or factorial when after a natural number (type-checking prevents ambiguity)
$\in$	is an element of (not to be confused with the Greek letter ϵ or ε)
$\implies$	implies
$\iff$	if and only if
$\cong$	Isomorphism, sometimes also denoted $\sim$ when it's a label on an arrow.
1_a	The identity morphism on an object a.
$\mathbb{1}$	Often denotes a category with one object and one morphism.
$g \circ f$ or gf	The composite of morphisms $\xrightarrow{f} \xrightarrow{g}$.
$\mathcal{C}(a, b)$	The collection (or set) morphisms from a to b in a category $\mathcal{C}$.
$\mathcal{C}^{\mathrm{op}}$	The dual category to $\mathcal{C}$.
$\longmapsto$	An arrow indicating the action of a function or functor on an individual element, or object/morphism.
$\hookrightarrow$	We often use this shape of arrow with a hook at the beginning to indicate some sort of inclusion, like the inclusion of a subset into the bigger set.

Further reading

I used the following two books to introduce undergraduate mathematicians to Category Theory for several years:

- Steve Awodey, *Category Theory*, volume 52 of Oxford Logic Guides, Oxford University Press, 2006.
- F. William Lawvere and Stephen H. Schanuel, *Conceptual Mathematics: A First Introduction to Categories*, Cambridge University Press, 1997

The following two excellent texts were published later; my students have found them harder than Awodey:

- Tom Leinster, *Basic Category Theory*, Cambridge Studies in Advanced Mathematics, Cambridge University Press, 2014.
- Emily Riehl, *Category Theory in Context*, Dover, 2016.

The following has an emphasis on applications (obviously):

- Brendan Fong and David I. Spivak, *An Invitation to Applied Category Theory: Seven Sketches in Compositionality*, Cambridge University Press, 2019.

The following are, as indicated by the titles, aimed at specific non-mathematician audiences:

- Bartosz Milewski, *Category Theory for Programmers*, available from github.com/hmemcpy/milewski-ctfp-pdf/ .
- David Spivak, *Category Theory for the Sciences*, MIT Press, 2014.

The following are the standard graduate-level texts on category theory.

- Saunders Mac Lane, *Categories for the Working Mathematician*, volume 5 of Graduate Texts in Mathematics, Springer-Verlag, second edition, 1998.
- Francis Borceux, *Handbook of Categorical Algebra 1 and 2*, volumes 50 and 51 of Encyclopedia of Mathematics and its Applications, Cambridge University Press, 1994.

The following is a semi-formal introduction to higher-dimensional category theory, with many pictures:

- Eugenia Cheng and Aaron Lauda, *Higher-dimensional Categories: An Illustrated Guidebook*, 2004, available from eugeniacheng.com/guidebook/ .

The following are other books by me which I referred to at various points (note that some of them have slightly different titles in the UK and the US):

- *How to Bake Pi: An Edible Exploration of the Mathematics of Mathematics*, Profile Books (Basic Books in the US), 2015.
- *Beyond Infinity: An Expedition to the Outer Limits of the Mathematical Universe*, Profile Books (Basic Books in the US), 2017.
- *The Art of Logic: How to Make Sense in a World that Doesn't*, Profile Books (Basic Books in the US), 2018.
- *X+Y: A Mathematician's Manifesto for Re-thinking Gender*, Profile Books (Basic Books in the US), 2020.

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Personal thanks with British spelling, as that's how I spell in normal life.

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I would also like to thank my undergraduate directors of studies Paul Glendinning and Jan Saxl, who encouraged me to keep going when, as a congressive student surrounded by ingressive and often over-confident (and usually male) peers, I very often felt sure I wasn't good enough. I am deeply saddened that Jan Saxl is no longer here to receive my thanks; I owe him a great deal, for always believing in me when I didn't. He not only accepted that I would sometimes neglect my mathematics to prepare for an upcoming concert, but he even came to the concerts. He entrusted me with teaching, much earlier than I expected, and went so smoothly from seeing me as a student to seeing me as a mathematician, that I was encouraged to see it myself.

Peter Johnstone not only taught me logic, which was my favourite undergraduate course, but he gave me my first chance to teach a course on Category Theory, which was in fact my first ever lecture course. Those notes are preserved in files typed up by Richard Garner, who was officially my student but I've definitely learnt more from him than whatever I taught him; similarly Nick Gurski, who has been my most fruitful and hilarious collaborator.

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